Chapter 7 Notes

Section 1

In the Remarks are page 91, it is said that there is an "easy induction" showing that R_{n+1} has 2^n elements. However, this is clearly not the case. For, as remakred, R_3 has 4 elements. Then R_4 has $2^4 = 16$ elements, which is not equal to $2^3 = 8$.

The proof of Proposition 1.3 is problematic. It uses Cantor's Theorem, though this hasn't even been stated (let alone proven) in the context of Zermelo-Frankel Universes. Proposition 1.3 can also be proven without using Cantor's Theorem. In particular, we first show that $R_\alpha \notin R_\alpha$.

Proof. By transfinite induction. Clearly this holds for 0. Suppose it holds for some α. Then, if $R_{\alpha+1} \in R_{\alpha+1}$, by definition of $R_{\alpha+1}$, $R_{\alpha+1}$ would be a subset of R_α. But $R_\alpha \in R_{\alpha+1}$. Thus this would mean that $R_\alpha \in R_\alpha$, contradicting the induction assumption. Hence $R_{\alpha+1} \notin R_{\alpha+1}$. On the other hand, suppose it held for all ordinals less than some limit ordinal λ. Then, by definition, $R_\lambda = \cup_{\alpha<\lambda} R_\alpha$. Suppose $R_\lambda \in R_\lambda$. Then, for some $\alpha < \lambda$, $R_\lambda \in R_\alpha$. But $R_{\alpha+1} \subset R_\lambda$, and $R_\alpha \in R_{\alpha+1}$, meaning $R_\alpha \in R_\lambda$. Thus, because R_α is transitive, $R_\alpha \in R_\alpha$, contradicting the induction assumption. Thus $R_\lambda \notin R_\lambda$, completing the transfinite induction. □

Then, in the proof of Proposition 1.3, $R_\alpha \neq R_{\alpha+1}$ follows for the following reason: We know that $R_\alpha \in R_{\alpha+1}$ and $R_\alpha \notin R_\alpha$ (by the above result), meaning $R_\alpha \neq R_{\alpha+1}$.

Section 2

Note that, at a minimum, the R_α sequence discussed in this section is an On-sequence of subsets of V (and of course, all On-sequences for any particular class are also On-sequences for V itself).

Note that the proof of O_6 uses Theorem 1.18 of Chapter 5.